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Inventor(s)	Ford; Brian C.

Hubcap

Abstract

A hubcap includes a monolithic main body. The monolithic main body includes (i) a sidewall having an annular cross-section, a hub end, and a vent end opposite the hub end, (ii) an end-cap abutting and spanning across the vent end, and (iii) a flange abutting and at least partially surrounding the hub end. The flange has an outer flange-surface facing the vent end and, opposite the outer flange-surface, an inner flange-surface that defines an annular channel.

Inventors:	Ford; Brian C. (Moseley, VA)
Applicant:	Amsted Rail Company, Inc. (Chicago, IL)
Family ID:	90000347
Assignee:	Amsted Rail Company, Inc. (Chicago, IL)
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Primary Examiner: Bellinger; Jason R

Attorney, Agent or Firm: COZEN O'CONNOR

Background/Summary

BACKGROUND

(1) Wheeled motor vehicles include wheel hub assemblies, each of which attaches a respective wheel to the vehicle's axle. A wheel hub assembly may include bearings, seals, the axle, an axle nut, and other components within a hub cavity. The bearings interact with the axle to enable the wheel to rotate smoothly with minimal friction. The hub cavity further decreases friction and dissipates heat by lubricating the bearings. A hubcap prevents contaminants from entering a hub assembly, while also protecting the hub from impacts and tampering.

SUMMARY OF THE EMBODIMENTS

(2) Typical hubcaps include a metal main body and a separate end-wall that covers a vent-end of the hubcap. The main body is metal to withstand impacts. The end-wall is transparent to enable viewing the lubricant chamber and oil level, which precludes the end-wall from being metal. Such a hubcap also includes a gasket between the main body and the end-wall, and may also include an additional gasket between the main body and the wheel hub. A common problem of many current hubcap designs is oil leakage at these gaskets. Embodiments of hubcaps disclosed herein remedy these problems. First, the hubcap's main body and end-wall are a single piece, such that there is no interface from which oil can leak. Second, the main body may include a sealing ring, which has one or more flanges to provide redundancy against leaks

(3) In a first aspect, a hubcap includes a monolithic main body. The monolithic main body includes (i) a sidewall having an annular cross-section, a hub end, and a vent end opposite the hub end, (ii) an end-cap abutting and spanning across the vent end, and (iii) a flange abutting and at least partially surrounding the hub end. The flange has an outer flange-surface facing the vent end and, opposite the outer flange-surface, an inner flange-surface that defines an annular channel.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIGS. 1-4 are respectively an isometric top view, an isometric bottom view, a first cross-sectional view, and a second cross-sectional view of a hubcap, in an embodiment.
- (2) FIGS. 5-7 is a cross-sectional view of a hubcap that includes an overmold, in which the hubcap is an embodiment of the hubcap of FIGS. 1-4.
- (3) FIG. 8 is a schematic of a sealing ring included in embodiments of hubcaps of FIGS. 1-7.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (4) FIGS. 1, 2, and 3 are respectively an isometric top view, an isometric bottom view, and a cross-sectional view of a hubcap **101**. Hubcap **101** includes a monolithic main body **100**, which has the following regions: a flange **110**, a sidewall **120**, and an end-cap **130**. FIG. 4 is a cross-sectional view of sidewall **120** in a cross-sectional plane 4-4' shown in FIG. 3. FIGS. 1-4 are best viewed together in the following description.
- (5) Flange **110**, sidewall **120**, and end-cap **130** may be monolithic, e.g., formed of a single piece of material, such as a single continuous volume of material. Flange **110**, sidewall **120**, and end-cap **130** may be integrally formed, such that they make up a single piece or unit, and cannot be dismantled without destroying the integrity of main body **100** or any of its components.
- (6) When flange **110**, sidewall **120**, and end-cap **130** are not monolithic, main body **100** may be integrally formed by irremovable attachment of at least one of flange **110**, sidewall **120**, and end-cap **130**, wherein, prior to said attachment, at least one of flange **110**, sidewall **120**, and end-cap **130** are separate pieces. For example, at least one of (a) flange **110** and sidewall **120** may be irremovably attached, and (b) sidewall **120** and end-cap **130** may be irremovably attached. The means of irremovable attachment may include welding, such as plastic welding, adhesive bonding, multi-part molding, or a combination thereof. At least two of flange **110**, sidewall **120**, and end-cap **130** may be formed of different respective materials. Main body **100** may include a volume of adhesive between at least one of at least one of (a) flange **110** and sidewall **120**, and (b) sidewall **120** and end-cap **130**.
- (7) Sidewall **120** has an annular cross-section, a hub end **122**, and a vent end **129** opposite hub end **122**. The annular cross-section of sidewall **120** may be a circular annulus, as shown in FIG. 4, or a non-circular annulus, such as a polygonal annulus. End-cap **130** abuts and spans across vent end **129**. In embodiments, sidewall **120** and end-cap **130** respectively form the lateral surface and a base surface of a frustum, such as a conical frustum or a polygonal frustum. Sidewall **120** has an interior surface **126** and an exterior surface **128**. End-cap **130** has an inner-facing surface **136** and an outer-facing surface **138**. The cross-sectional plane of FIG. 3 is perpendicular to at least one of surfaces **136** and **138**, and may bisect hubcap **101**.
- (8) Flange **110** abuts and at least partially surrounds hub end **122**. Flange **110** has an outer flange-surface **118** facing vent end **129**, and opposite outer flange-surface **118**, an inner flange-surface **112** that defines an annular channel **116**. Specifically, inner flange-surface **112** includes a recessed region **113** that defines annular channel **116**. In embodiments, inner flange-surface **112** also forms an annular channel **114**, which is surrounded by annular channel **116**.
- (9) As shown in FIG. 2, monolithic main body **100** may include a plurality of communication holes **119**, each spanning between recessed region **113** and one of outer flange-surface **118** (as shown in FIG. 3) and exterior surface **128**. FIG. 3 illustrates a communication hole **319**, which is an example of communication hole **119** that is coplanar with a bolt-hole **117** in the cross-sectional plane of FIG. 3. Communication holes **119** may be coplanar with a bolt-hole **117**, as in FIG. 3, or not coplanar in this sense, as illustrated by communication holes **119** of FIG. 2. For compatibility with tooling used in manufacturing main body **100**, an axis of communication hole **319** may be at least one of (a) perpendicular to one or more of surfaces **112**, **136**, and **138**, (b) parallel to an axis of a bolt-hole

117, and (c) parallel to a symmetry axis of main body **100**.

(10) Each communication hole **119** may be defined by a respective surface **111** of main body **100**, which spans between recessed region **113** and one of outer flange-surface **118** (as shown in FIG. 3) and exterior surface **128**. The total number of communication holes **119** of main body **100** may be between two and eight, inclusive. For example, FIG. 2 denotes three communication holes **119**.

(11) Inner flange-surface **112** may be a bottom surface of hubcap **101**, such that when hubcap **101** is installed on a hub assembly, at least part of inner flange-surface **112** is in direct contact with one of (i) the hub assembly and (ii) a gasket or sealing ring disposed between hubcap **101** and the hub assembly. For example, hubcap **101** may include at least one of a sealing ring **316** located at least partially in annular channel **116**, and a sealing ring **314** located at least partially in annular channel **114**. Each of sealing rings **314** and **316** contribute forming a seal at the interface of inner flange-surface **112** and a surface of a hub assembly to which hubcap **101** is attached. In embodiments, hubcap **101** includes gasket, such as a flange gasket, removably attached to inner flange-surface **112**. This gasket may function as at least one of sealing ring **314** and **316**, such that part of the gasket is in at least one of annular channel **114** and **116**.

(12) Monolithic main body **100** may be formed a high-impact plastic, examples of which include polyethylene terephthalate (PET) and polycarbonate. The plastic may have an Izod Impact Strength of at least 860 joules per meter. Flange **110**, sidewall **120**, and end-cap **130** have respective thicknesses **115**, **125**, and **135**, each of which may be between 4.5 millimeters and 8.0 millimeters. Monolithic main body **100** may be either opaque or transparent to visible light. Transparency of monolithic main body **100** enables visual inspection of oil levels in a hub assembly to which hubcap **101** is attached.

(13) In embodiments, flange **110** includes a plurality of bolt-holes **117** for bolting main body **100** to a hub assembly. Flange **110** may also include a respective compression limiter in each bolt-hole **117** to prevent main body **100** from cracking when bolts through bolt-holes **117** are tightened.

(14) In embodiments, hubcap **101** includes at least one of a vent **134** on outer-facing surface **138**, and a filter **132** adjacent to outer-facing surface **138**. At least one of filter **132** and vent **334** may be removably attached to end-cap **130**. Vent **134** may be fixedly attached to end-cap **130**. End-cap **130** may also include, between surfaces **136** and **138**, a surface **137** that defines an aperture through end-cap **130**.

(15) FIG. 5 is a cross-sectional view of a hubcap **501**, which is an embodiment of hubcap **101**. FIG. 6 is an isometric top view of hubcap **501**. Hubcap **501** includes an overmold **500** and main body **100**, which includes outer flange-surface **118** and exterior surface **128** of sidewall **120**. Overmold **500** covers, and is conformal to, at least part of each of (i) the outer flange-surface **118** and exterior surface **128**. Overmold **500** may also cover, and be conformal to, at least part of outer-facing surface **138** of end-cap **130**. Overmold **500** may be formed of a thermoplastic elastomer, which increases hubcap **501**'s impact resistance. Overmold **500** also enables hubcap **501** to satisfies market demand for hubcaps that are at least partially opaque.

(16) In embodiments, hubcap **101** includes a sealing ring **516** in annular channel **116**. Sealing ring **516** is an example of sealing ring **316**. For example, hubcap **501** may include sealing ring **516**, as shown in FIG. 5. Sealing ring **516** may be bonded to recessed region **113**. Sealing ring **516** may be either unconnected to overmold **500**, or connected to overmold **500**.

(17) A cross-sectional shape of sealing ring **516** may be that of a convex shape, such as a circle, ellipse, or a convex polygon. Alternatively, the cross-sectional shape may be that of convex shape, e.g., one with multiple lobes **517** as shown in FIG. 5. Lobes **517** on provide redundancy against leaks.

(18) When sealing ring **516** is connected to overmold **500**, hubcap **501** may include a plurality of spokes **518**, each of which is formed of the same material as at least one of overmold **500** and sealing ring **516**. In embodiments, spokes **518** may be conformally attached to both sealing ring **516** and overmold **500**. Each spoke **518** (a) extends through a respective communication hole **119**,

and (b) connecting respective region of sealing ring **516** with a respective region of overmold **500** that covers the respective communication hole **119**. In such embodiments, sealing ring **516** and overmold **500** may be monolithic, that is, they are formed of a continuous piece of material.

(19) Spokes **518** may be formed in the same process step as forming overmold **500** on main body **100**. In such embodiments, spokes **518** and overmold **500** are monolithic, and spokes **518** may be conformally attached to sealing ring **516**. Sealing ring may also be formed in the same process step as spokes **518** and overmold **500**. In such embodiments, overmold **500**, spokes **518**, and sealing ring **516** may be monolithic

(20) Forming sealing ring **516** and overmold **500** during a same process step eliminates material and labor costs associated with a separate sealing ring. Sealing ring **516**'s being attached to overmold **500** also facilitates removal and reinstallation of hubcap **501**, as there is no separate step of installing a separate ring, during which the separate ring might be damaged.

(21) Each communication hole **119** may have a hydraulic diameter $D_{sub.h}$ between 1.0 and 1.5 millimeters, inclusive. Hydraulic diameter $D_{sub.h}$ may be defined as $D_{sub.h} = 4A/P$, where A and P are, respectively, the cross-sectional area and total perimeter of communication hole **119**. When the hydraulic diameter exceeds one millimeter, communication holes **119** are sufficiently large to enable material of overmold **500** to flow through communication holes **119** and adjoin sealing ring **516**. When the hydraulic diameter is less than 1.5 millimeter, communication holes **119** are sufficiently small that they do not interfere with bolt-holes **117**.

(22) FIG. 7 is a cross-sectional view of hubcap **501**, in a cross-sectional plane that does not include a bolt-hole **117**. In such a cross-sectional plane, hubcap **501** may include a communication hole **719** and a spoke **718** therein. Communication hole **719** and spoke **718** are respective examples of communication hole **119** and spoke **518**.

(23) FIG. 8 is a schematic of a sealing ring **816** and three spokes **818**. Sealing ring **816** and spokes **818** may be a single piece of material, and may be part of a monolithic volume of material that includes overmold **500**. Sealing ring **816** is an example of sealing ring **516**. Each spoke **818** is an example of spokes **518** and **718**. In embodiments, sealing ring **816** is in annular channel **116**, and each spoke **818** is in a respective communication hole **119** shown in FIG. 1.

(24) Combinations of Features

(25) Features described above as well as those claimed below may be combined in various ways without departing from the scope hereof. The following enumerated examples illustrate some possible, non-limiting combinations:

(26) (A1) A hubcap includes: a monolithic main body including (i) a sidewall having an annular cross-section, a hub end, and a vent end opposite the hub end, (ii) an end-cap abutting and spanning across the vent end, and (iii) a flange abutting and at least partially surrounding the hub end. The flange has an outer flange-surface facing the vent end and, opposite the outer flange-surface, an inner flange-surface that defines an annular channel.

(27) (A2) Embodiments of hubcap (A1) further include an overmold covering, and conformal to, at least part of each of (i) the outer flange-surface and (ii) an exterior surface of the sidewall.

(28) (A3) In embodiments of hubcap (A2), the overmold also covers, and is conformal to, at least part of the end-cap.

(29) (A4) In embodiments of either of hubcaps (A2) and (A3), the overmold is formed of a thermoplastic elastomer.

(30) (A5) Embodiments of any of hubcaps (A2)-(A4) further include a sealing ring disposed in the annular channel.

(31) (A6) In embodiments hubcap (A5), the sealing ring and the overmold are unconnected.

(32) (A7) In embodiments of either of hubcaps (A5) and (A6), the inner flange-surface includes a recessed region that defines the annular channel, and the sealing ring is bonded to the recessed region.

(33) (A8) In embodiments of any of hubcaps (A5)-(A7), the inner flange-surface includes a

recessed region that defines the annular channel, and the monolithic main body includes a plurality of communication holes each spanning between the recessed region and a surface of the hubcap covered by the overmold. In such embodiments, the hubcap further includes a plurality of spokes. Each of the plurality of spokes (a) extends through a respective communication hole of the plurality of communication holes, and (b) connects a respective region of the sealing ring with a respective region of the overmold that covers the respective communication hole.

(34) (A9) In embodiments of hubcap (A8), a number of the plurality of communication holes being between two and eight.

(35) (A10) In embodiments of either of hubcaps (A8) and (A9), each of the plurality of communication holes having a hydraulic diameter between 1.0 millimeters and 1.5 millimeters.

(36) (A11) In embodiments of any of hubcaps (A8)-(A10), the inner flange-surface includes a recessed region that defines the annular channel, each of the plurality of communication holes each spanning between the recessed region and one of (i) the outer flange-surface and (ii) a surface of the sidewall.

(37) (A12) In embodiments of any of hubcaps (A8)-(A11), the main body has a plurality of interior surfaces, each of the plurality of interior surfaces (i) defining a respective one of the plurality of communication holes and (ii) spanning between the recessed region and a surface of the hubcap covered by the overmold.

(38) (A13) In embodiments of any of hubcaps (A1)-(A12), the main body is formed of a high-impact plastic.

(39) (A14) In embodiments of any of hubcaps (A1)-(A13), the main body is transparent to visible light.

(40) (A15) In embodiments of any of hubcaps (A1)-(A13), the main body is opaque to visible light.

(41) (A16) In embodiments of any of hubcaps (A1)-(A15), a thickness of the main body is between 4.5 mm and 8.0 mm.

(42) (A17) In embodiments of any of hubcaps (A1)-(A16), the monolithic main body has an Izod Impact Strength of at least 860 joules per meter.

(43) (A18) In embodiments of any of hubcaps (A1)-(A17), the inner flange-surface is a bottom surface of the hubcap.

(44) (A19) Embodiments of any of hubcaps (A1)-(A18) further include a vent attached to an outer-facing surface of the end-cap; and a filter removably attached in an inner-facing surface of the end-cap.

(45) Changes may be made in the above methods and systems without departing from the scope of the present embodiments. It should thus be noted that the matter contained in the above description or shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. Herein, and unless otherwise indicated, the phrase “in embodiments” is equivalent to the phrase “in certain embodiments,” and does not refer to all embodiments. The following claims are intended to cover all generic and specific features described herein, as well as all statements of the scope of the present method and system, which, as a matter of language, might be said to fall therebetween.

Claims

1. A hubcap comprising: a monolithic main body including (i) a sidewall having an annular cross-section, a hub end, and a vent end opposite the hub end, (ii) an end-cap abutting and spanning across the vent end, and (iii) a flange abutting and at least partially surrounding the hub end; the flange having an outer flange-surface facing the vent end and, opposite the outer flange-surface, an inner flange-surface that defines an annular channel; and the monolithic main body including a plurality of communication holes each spanning between the annular channel and the outer flange-surface.

2. The hubcap of claim 1, further comprising an overmold covering, and conformal to, at least part of each of (i) the outer flange-surface and (ii) an exterior surface of the sidewall.
 3. The hubcap of claim 2, the overmold also covering, and conformal to, at least part of the end-cap.
 4. The hubcap of claim 2, the overmold comprising a thermoplastic elastomer.
 5. The hubcap of claim 2, further comprising a sealing ring disposed in the annular channel.
 6. The hubcap of claim 5, the sealing ring and the overmold being unconnected.
 7. The hubcap of claim 5, the inner flange-surface including a recessed region that defines the annular channel, the sealing ring being bonded to the recessed region.
 8. The hubcap of claim 5, the inner flange-surface including a recessed region that defines the annular channel, the hubcap further comprising a plurality of spokes, each of the plurality of spokes (a) extending through a respective communication hole of the plurality of communication holes, and (b) connecting a respective region of the sealing ring with a respective region of the overmold that covers the respective communication hole.
 9. The hubcap of claim 8, a number of the plurality of communication holes being between two and eight.
 10. The hubcap of claim 8, each of the plurality of communication holes having a hydraulic diameter between 1.0 millimeters and 1.5 millimeters.
 11. The hubcap of claim 8, the inner flange-surface including a recessed region that defines the annular channel, each of the plurality of communication holes each spanning between the recessed region and one of (i) the outer flange-surface and (ii) a surface of the sidewall.
 12. The hubcap of claim 8, the main body having a plurality of interior surfaces, each of the plurality of interior surfaces (i) defining a respective one of the plurality of communication holes and (ii) spanning between the recessed region and a surface of the hubcap covered by the overmold.
 13. The hubcap of claim 1, the main body being formed of a high-impact plastic.
 14. The hubcap of claim 1, the main body being transparent to visible light.
 15. The hubcap of claim 1, the main body being opaque to visible light.
 16. The hubcap of claim 1, a thickness of the main body being between 4.5 mm and 8.0 mm.
 17. The hubcap of claim 1, the monolithic main body having an Izod Impact Strength of at least 860 joules per meter.
 18. The hubcap of claim 1, the inner flange-surface being a bottom surface of the hubcap.
 19. The hubcap of claim 1, further comprising: a vent attached to an outer-facing surface of the end-cap; and a filter removably attached in an inner-facing surface of the end-cap.
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